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tsspiral: An R Package to Create Spiral Version of Time Series Plots

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Summary

Time series data commonly display seasonal over time. Circular visualisation provides a natural way to represent these recurring patterns. Since most statistical software packages readily support line plots, their convenience and widespread availability have made line plots the conventional choice for visualising time series. As a result, other forms of visualisation have received comparatively less attention, despite their potential to reveal recurring and circular patterns in time series data. tsspiral is an open-source R software for visualising time series on a polar coordinate.

By bringing polar-coordinate-based visualisation into routine time series analysis, tsspiral offers a practical tool for exploring temporal patterns and communicating complex time-dependent structures.

Statement of need

tsspiral beridges these gaps. It aims to overcomes previous limitations, while adding support for

With conventional time series plots it is very difficult to nested temporal structures, compare seasonal patterns across different time scales, identify missing values,particularly in long time series specially when you have daily data. Spiral vrsion of time series plot are useful in addressing these limitations. The spiral plots supports visualizing long time series and better identification of periodic structures in the data (Rakuschek, Hauser, & Schreck, 2026; Weber, Alexa, & Müller, 2001). As John W. Tukey emphasised, "Exploratory data analysis is an attitude, a state of flexibility, a willingness to look for those things that we believe are not there, as well as those we believe to be there."(@ Tukey et al., 1977) Guided by this philosophy, exploring alternative visualisations of time series data encourages us to move beyond familiar representations and discover patterns that might otherwise remain hidden. Creating spiral time series plot often reqires custom codes and careful handing of time series data. Furthermore, when data comes as categorical variables we can not use traditional time series plots to visualise data.

Here is an motivation example. The following graph shows relatil bean price and Dambulla vegetable market in Sri Lanka.

```
library(tsspiral)
library(viridis)
library(ggplot2)
library(tsibble)
library(tidyverse)
data(tourism.srilanka)
ts <- tourism.srilanka |>
  mutate(Date = dmy(Date)) |>
```